

Course 5092

5 Credits: 4

Program Core

Source-grounded

Quality Assurance &Testing

□ To understand the significance in quality control of polymer products.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5092 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5092.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	5092
Course title	Quality Assurance &Testing
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To understand the significance in quality control of polymer products.
- To impart knowledge on raw material testing, processability tests and product testing which comes under both rubbers and plastics.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding and, electrical properties of polymers. Summarize test methods for ISNR and concentrated	See official syllabus
CO2	14 Understanding latex as per IS standards Explain the testing procedures for physical	See official syllabus
CO3	15 Understanding properties of polymers. Explain the specific testing procedures for dry	See official syllabus
CO4	14 Understanding rubber, plastic and latex products	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	properties of polymers	4	As specified
Module 2	standards	4	As specified
Module 3	Explain the testing procedures for physical properties of polymers	4	As specified
Module 4	products	4	As specified

MODULE 1

properties of polymers

This module is presented from the official syllabus scope for Quality Assurance & Testing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: properties of polymers
- Official duration: As specified

- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

1 Remembering of standards organizations Summarize the role of standards

1 Remembering of standards organizations Summarize the role of standards is an official topic in Module 1 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Remembering of standards organizations Summarize the role of standards using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 remembering of standards organizations summarize the role of standards should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Remembering of standards organizations Summarize the role of standards means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1- 1 Remembering of standards organizations
Summarize the role of standards organization definition/meaning
organization. organization organization, steps, application organization
organization. Technical terms English- organization organization.

organizations in standardization and quality 3 Understanding assurance. Explain test procedures for purity of raw

organizations in standardization and quality 3 Understanding assurance. Explain test procedures for purity of raw is an official topic in Module 1 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify organizations in standardization and quality 3 Understanding assurance. Explain test procedures for purity of raw using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, organizations in standardization and quality 3 understanding assurance. explain test procedures for purity of raw should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What organizations in standardization and quality 3 Understanding assurance. Explain test procedures for purity of raw means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓർഗനൈസേഷനുകളും സ്റ്റാൻഡറൈസേഷനും ക്വാളിറ്റി അസുറൻസ്. Explain test procedures for purity of raw materials. ഓർഗനൈസേഷനുകളും സ്റ്റാൻഡറൈസേഷനും ക്വാളിറ്റി അസുറൻസ്, steps, application ഓർഗനൈസേഷനുകളും സ്റ്റാൻഡറൈസേഷനും ക്വാളിറ്റി അസുറൻസ്. Technical terms English-ൽ എഴുതുക. ഓർഗനൈസേഷനുകളും സ്റ്റാൻഡറൈസേഷനും ക്വാളിറ്റി അസുറൻസ്.

4 Understanding materials Describe test methods for thermal and

4 Understanding materials Describe test methods for thermal and is an official topic in Module 1 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding materials Describe test methods for thermal and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding materials describe test methods for thermal and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding materials Describe test methods for thermal and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-4 Understanding materials Describe test methods for thermal and mechanical properties. **English:** definition/meaning, steps, application, technical terms. **English:** definition/meaning, steps, application, technical terms.

7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS

7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS is an official topic in Module 1 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical

properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 understanding electrical properties of polymers. importance of quality assurance and testing-quality-definition. modern quality concepts, importance of quality control in polymer industry. standards and specifications. different standard organizations - iso, astm, bis, din. procedure for accreditation of isi certification. role of standard organization in quality assurance and customer satisfaction. raw material testing - testing of additives-% purity of dpg, mbt, zno. thermal properties - thermal conductivity, flammability, vicat softening temperature, melting point, brittleness temperature, specific heat, hdt. electrical properties - dielectric strength, dielectric constant, arc resistance. summarize test methods for isnr and concentrated latex as per is should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS definition/meaning. steps, application. Technical terms English-Malayalam.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

properties of polymers		
Outline the quality concepts and importance		
M1.01		1
Remembering		
of standards organizations		
Summarize the role of standards		
M1.02		3
Understanding		
organizations in standardization and quality		
assurance.		
Explain test procedures for purity of raw		
M1.03		4
Understanding		
materials		
Describe test methods for thermal and		
M1.04		7
Understanding		
electrical properties of polymers.		

Contents:

Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

standards

This module is presented from the official syllabus scope for Quality Assurance & Testing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: standards
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR

procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR is an official topic in Module 2 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, procedure to determine tests of dry rubber 6 understanding (isnr). compile the technical specifications of isnr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രക്രിയ നിർണ്ണയിക്കുന്നതിനുള്ള ഡ്രൈ റബ്ബർ 6 Understanding (ISNR). Compile the technical specifications of ISNR പദപരിചയം/പദാർത്ഥം definition/meaning പദപരിചയം/പദാർത്ഥം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം പദപരിചയം/പദാർത്ഥം.

1 Understanding as per IS 4588. Explain the significance, principle and

1 Understanding as per IS 4588. Explain the significance, principle and is an official topic in Module 2 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding as per IS 4588. Explain the significance, principle and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding as per is 4588. explain the significance, principle and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding as per IS 4588. Explain the significance, principle and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- 1 Understanding as per IS 4588. Explain the significance, principle and definition/meaning.
 steps, application. Technical terms English-

procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of

procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of is an official topic in Module 2 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, procedure to determine tests of concentrated 6 understanding latex. compile the technical specifications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രക്രിയ നിർണ്ണയിക്കുന്നതിനുള്ള 6 ഉൾക്കൊള്ളിയിൽ. Compile the technical specifications of ഏതെങ്കിലും പദം definition/meaning പദം. പദം പദം പദം, steps, application പദം പദം പദം. Technical terms English-ൽ പദം പദം.

1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile mater, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity

1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile mater, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity is an official topic in Module 2 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile mater, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile matter, ash content, N₂ content, Po and PRI, ash and colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for concentrated latex –TSC, DRC, KOH number, VFA number, mechanical stability time, Mg content, sludge and coagulum content, alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for viscosity and specific gravity should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile matter, ash content, N ₂ content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile matter, ash content, N₂ content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity definition/meaning. Steps, application. Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

standards

M2.01	Explain the significance, principle and procedure to determine tests of dry rubber	6
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Understanding

(ISNR).

Compile the technical specifications of ISNR

M2.02		1
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Understanding

as per IS 4588.

Explain the significance, principle and

M2.03	procedure to determine tests of concentrated	6
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Understanding

latex.

Compile the technical specifications of

M2.04		1
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Understanding

concentrated latex as per IS 3708.

	Series Test I	1
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Contents:

Significance, principle and method of tests for ISNR – dirt content, volatile mater, ash

content, N₂ content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS.

Significance, principle and methods of tests for Concentrated latex –TSC, DRC,

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain the testing procedures for physical properties of polymers

This module is presented from the official syllabus scope for Quality Assurance & Testing. Use the topic cards for learning and the source transcript for exact

coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the testing procedures for physical properties of polymers
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

4 Understanding properties of plastics and rubbers Describe the test method to determine

4 Understanding properties of plastics and rubbers Describe the test method to determine is an official topic in Module 3 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding properties of plastics and rubbers Describe the test method to determine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding properties of plastics and rubbers describe the test method to determine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding properties of plastics and rubbers Describe the test method to determine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Understanding properties of plastics and rubbers Describe the test method to determine ΔT_g definition/meaning ΔT_g . ΔT_g ΔT_g ΔT_g , steps, application ΔT_g ΔT_g ΔT_g . Technical terms English- ΔT_g ΔT_g ΔT_g .

4 Understanding permanent set of rubbers. Describe the test method to determine

4 Understanding permanent set of rubbers. Describe the test method to determine is an official topic in Module 3 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding permanent set of rubbers. Describe the test method to determine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding permanent set of rubbers. describe the test method to determine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding permanent set of rubbers. Describe the test method to determine means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Understanding permanent set of rubbers. Describe the test method to determine ΔL definition/meaning ΔL . ΔL ΔL ΔL , steps, application ΔL ΔL . Technical terms English- ΔL ΔL ΔL .

hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of

hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of is an official topic in Module 3 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, hardness, abrasion resistance, resilience, flex 6 understanding crack, fatigue failure and impact testing. outline the procedure to determine mfi of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഘർഷണം, ഘർഷണ പ്രതിരോധം, ഘർഷണ പ്രതിരോധം, ഘർഷണ പ്രതിരോധം 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of ഘർഷണ പ്രതിരോധം definition/meaning ഘർഷണ പ്രതിരോധം. ഘർഷണ പ്രതിരോധം, steps, application ഘർഷണ പ്രതിരോധം ഘർഷണ. Technical terms English-ഘർഷണ പ്രതിരോധം.

1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers - Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex

1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers - Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests - Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex is an official topic in Module 3 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers - Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests - Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 understanding plastics. method of testing, principle of conditioning. significance, specimen standards, test methods relevant to plastics and

rubbers – tensile strength, m300 , elongation at break, tear strength, tension set, compression set, hardness, abrasion resistance, resilience, flex cracking, fatigue failure, impact testing of plastics- izod and charpy processability tests -determination of melt flow index of plastics. explain the specific testing procedures for dry rubber, plastic and latex should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers – Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers – Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Explain the testing procedures for physical properties of polymers

Describe the test method to determine tensile

M3.01

4

Understanding

properties of plastics and rubbers

Describe the test method to determine

M3.02

4

Understanding

permanent set of rubbers.

Describe the test method to determine

M 3.03

hardness, abrasion resistance, resilience, flex

6

Understanding

crack, fatigue failure and impact testing.

Outline the procedure to determine MFI of

M3.04

1

Understanding

plastics.

Contents:

Method of testing, principle of conditioning. Significance, specimen standards, test

methods relevant to plastics and rubbers – Tensile strength, M300 , Elongation at break, Tear

strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex

cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

products

This module is presented from the official syllabus scope for Quality Assurance & Testing. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: products
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

State the importance of product testing. 1 Remembering Outline the test procedure and specifications

State the importance of product testing. 1 Remembering Outline the test procedure and specifications is an official topic in Module 4 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify State the importance of product testing. 1 Remembering Outline the test procedure and specifications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, state the importance of product testing. 1 remembering outline the test procedure and specifications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What State the importance of product testing. 1 Remembering Outline the test procedure and specifications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓൺ State the importance of product testing. 1 Remembering Outline the test procedure and specifications ഓൺഓൺഓൺഓൺഓൺഓൺ definition/meaning ഓൺഓൺഓൺഓൺഓൺ. ഓൺഓൺഓൺ ഓൺഓൺഓൺ, steps, application ഓൺഓൺഓൺ ഓൺഓൺഓൺ ഓൺഓൺ. Technical terms English-ഓൺ ഓൺഓൺ ഓൺഓൺഓൺഓൺ.

5 Understanding for dry rubber products. Explain the test procedure and specifications

5 Understanding for dry rubber products. Explain the test procedure and specifications is an official topic in Module 4 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding for dry rubber products. Explain the test procedure and specifications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding for dry rubber products. explain the test procedure and specifications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding for dry rubber products. Explain the test procedure and specifications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 5 Understanding for dry rubber products.

Explain the test procedure and specifications **பரீட்சை நடைமுறை** definition/meaning **பரீட்சை விவரம்**. **பரீட்சை** **பரீட்சை** **பரீட்சை**, steps, application **பரீட்சை** **பரீட்சை** **பரீட்சை**. **Technical terms** English-**பரீட்சை** **பரீட்சை**.

4 Understanding for latex products. Explain the test procedure and specifications

4 Understanding for latex products. Explain the test procedure and specifications is an official topic in Module 4 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding for latex products. Explain the test procedure and specifications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding for latex products. explain the test procedure and specifications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding for latex products. Explain the test procedure and specifications means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-4 Understanding for latex products. Explain the test procedure and specifications ഡിഫിനിഷൻ/മിനിമം definition/meaning ഡിഫിനിഷൻ/മിനിമം. ഡിഫിനിഷൻ ഡിഫിനിഷൻ, steps, application ഡിഫിനിഷൻ ഡിഫിനിഷൻ. Technical terms English-4 ഡിഫിനിഷൻ ഡിഫിനിഷൻ.

4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers - Scott R1 Physical testing of Rubbers - R.P. Brown R2 Indian Standards R3 ASTM standards - Vol. 8 & 9 series R4 Plastic Testing Technology - Vishu Shah Online Resources: Sl.No Website Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com>

4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers - Scott R1 Physical testing of Rubbers - R.P. Brown R2 Indian Standards R3 ASTM standards - Vol. 8 & 9 series R4 Plastic Testing Technology - Vishu Shah Online Resources: Sl.No Website Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com> is an official topic in Module 4 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers - Scott R1 Physical testing of Rubbers - R.P. Brown R2 Indian Standards R3 ASTM standards - Vol. 8 & 9 series R4 Plastic Testing Technology - Vishu Shah Online

Resources: Sl.No Website Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com> using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding for plastic products. importance of product testing, testing of dry rubber products- is specifications and specific test procedures of cycle tyre, cycle tube, m.c. sheet, v-straps. testing of latex products- is specifications and specific test procedures of latex foam, surgical gloves and examination gloves. testing of plastic product - pvc pipe, water tank, helmet text / reference: t/r book title/author t1 physical testing of rubbers – scott r1 physical testing of rubbers – r.p. brown r2 indian standards r3 astm standards – vol. 8 & 9 series r4 plastic testing technology – vishu shah online resources: sl.no website link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers – Scott R1 Physical testing of Rubbers – R.P. Brown R2 Indian Standards R3 ASTM standards – Vol. 8 & 9 series R4 Plastic Testing Technology – Vishu Shah Online Resources: Sl.No Website Link 1 https://www.astm.org 2 https://www.astm.org 3 https://link.springer.com 4 http://www.intertek.com 5 https://www.smithersrapra.com means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Understanding for plastic products.

Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers – Scott R1 Physical testing of Rubbers – R.P. Brown R2 Indian Standards R3 ASTM standards – Vol. 8 & 9 series R4 Plastic Testing Technology – Vishu Shah Online Resources: Sl.No Website Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com> definition/meaning steps, application Technical terms English-

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

products		
M4.01	State the importance of product testing.	1
Remembering		
	Outline the test procedure and specifications	
M4.02		5
Understanding		
	for dry rubber products.	
	Explain the test procedure and specifications	
M4.03		4
Understanding		
	for latex products.	
	Explain the test procedure and specifications	
M4.04		4
Understanding		
	for plastic products.	
	Series Test II	1
Contents:		
Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps.		
Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves.		
Testing of Plastic product - PVC pipe, Water tank, Helmet		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 1 Remembering of standards organizations Summarize the role of standards. (3 marks)

Answer: Begin with the standard definition or identity of *1 Remembering of standards organizations Summarize the role of standards*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain organizations in standardization and quality 3

Understanding assurance. Explain test procedures for purity of raw. (3 marks)

Answer: Begin with the standard definition or identity of *organizations in standardization and quality 3 Understanding assurance. Explain test procedures for purity of raw*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding materials Describe test methods for thermal and. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding materials Describe test methods for thermal and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS. (3 marks)

Answer: Begin with the standard definition or identity of 7 *Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.*

Malayalam support: ഏകീകൃത നിർവ്വചനം, പ്രവർത്തനം, ഉപയോഗം എന്നിവ ഉൾക്കൊള്ളുന്ന മലയാളം വിവരണം.

Module 2

Define or explain procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR. (3 marks)

Answer: Begin with the standard definition or identity of *procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഏകീകൃത നിർവ്വചനം, പ്രവർത്തനം, ഉപയോഗം എന്നിവ ഉൾക്കൊള്ളുന്ന മലയാളം വിവരണം.

Define or explain 1 Understanding as per IS 4588. Explain the significance, principle and. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding as per IS 4588. Explain the significance, principle and.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of. (3 marks)

Answer: Begin with the standard definition or identity of *procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR - dirt content, volatile mater, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex -TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR - dirt content, volatile mater, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex -TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 3

Define or explain 4 Understanding properties of plastics and rubbers Describe the test method to determine. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding properties of plastics and rubbers Describe the test method to determine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 4 Understanding permanent set of rubbers. Describe the test method to determine. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding permanent set of rubbers. Describe the test method to determine*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of. (3 marks)

Answer: Begin with the standard definition or identity of *hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers - Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex. (3 marks)

Answer: Begin with the standard definition or identity of *1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers - Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain State the importance of product testing. 1 Remembering Outline the test procedure and specifications. (3 marks)

Answer: Begin with the standard definition or identity of *State the importance of product testing. 1 Remembering Outline the test procedure and specifications.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 5 Understanding for dry rubber products. Explain the test procedure and specifications. (3 marks)

Answer: Begin with the standard definition or identity of 5 *Understanding for dry rubber products*. Explain the test procedure and specifications. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഏകദേശം definition, തത്വം principle/steps, പ്രയോഗം application എന്നിവ ഉൾപ്പെടെ.

Define or explain 4 Understanding for latex products. Explain the test procedure and specifications. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding for latex products*. Explain the test procedure and specifications. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: ഏകദേശം definition, തത്വം principle/steps, പ്രയോഗം application എന്നിവ ഉൾപ്പെടെ.

Define or explain 4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers - Scott R1 Physical testing of Rubbers - R.P. Brown R2 Indian Standards R3 ASTM standards - Vol. 8 & 9 series R4 Plastic Testing Technology - Vishu Shah Online Resources: Sl.No Website Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com>. (3 marks)

Answer: Begin with the standard definition or identity of 4 *Understanding for plastic products*. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers - Scott R1 Physical testing of Rubbers - R.P. Brown R2 Indian Standards R3 ASTM standards - Vol. 8

& 9 series R4 Plastic Testing Technology – Vishu Shah Online Resources: Sl.No Website
Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4
<http://www.intertek.com> 5 <https://www.smithersrapra.com>. State the governing
principle, list the key components or stages, and close with one relevant application or
observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,
□□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **1 Remembering of standards organizations Summarize the role of standards.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **4 Understanding properties of plastics and rubbers Describe the test method to determine.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **State the importance of product testing. 1 Remembering Outline the test procedure and specifications.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.astm.org> <https://www.astm.org> <https://link.springer.com>
<http://www.intertek.com> <https://www.smithersrapra.com>

IN-PAGE SEARCH

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